

UNIT - 1

1. List and explain the Java buzzwords.
2. Write all the Operators in Java and classify them with suitable example programs.
3. Explain the Object-Oriented Programming (OOP) paradigm. Discuss how the concepts of abstraction, encapsulation, inheritance and polymorphism are supported in Java with suitable examples.
4. What is constructor overloading and method overloading in Java? Write a Java program to illustrate both, along with the rules of parameter passing.
5. Describe a Constructor. What are its types? Explain constructor overloading with suitable example.
6. Explain Static keyword (variables, methods and block) with suitable programs.
7. A program is required to perform addition, subtraction, multiplication, division and modulus on two integer values. Identify the Java concept involved and write the program using a = 12 and b = 5. [Arithmetic operations (a=12, b=5)]
8. An object should be initialized automatically when it is created, without passing any values to it. Identify the type of constructor involved and write the program using the Student class to display "05 ASDF". [Default constructor – Student class]
9. a) A program needs to check whether two given values are equal, greater or smaller and use the result for a decision. Identify the Java concept involved and write a small program using a = 10 and b = 5 to demonstrate it. [Relational operators (a=10,b=5)]
b) An object needs to receive three values when it is created, and those values should be displayed through a method. Identify the type of constructor involved and write a small program using demonstration(1,2,3). [parameterized Constructor]
10. a) A message is common to all objects of a class and should be displayed without creating an object. Identify the Java feature involved and write a small program using the Demo class to display "Welcome to Java". (a) Static method/message (Demo class)

UNIT - 2

- 1) a) Explain Inheritance in Java and write the types of inheritance with suitable example programs.
b) What are member access rules in Inheritance? Explain the use of super and final keywords with examples.
- 2) a) What is Polymorphism? Explain Method Overriding with a suitable program.
b) Explain Abstract Classes and Object class in Java.
- 3) a) Explain Packages in Java. How to access, create and import packages?
b) What are Interfaces in Java? Explain implementing and extending interfaces with suitable examples.
- 4) Explain the different forms of inheritance supported by Java. Write a Java program to demonstrate multilevel and hierarchical inheritance using the super keyword.
- 5) Differentiate between abstract classes and interfaces in Java. Explain, with a program, the situations in which each should be used.
- 6) Explain the concept of packages in Java. Describe the steps involved in creating, compiling and accessing a user-defined package, with an example.
- 7) A new class needs to reuse the eat() behavior of an existing class and add its own bark() behavior. Identify the type of inheritance involved and write the program using Animal and Dog. [Inheritance – Animal/Dog (eat/bark)]
- 8) A child class needs to provide its own behavior for a method already available in its parent class. Identify the polymorphism concept involved and write the program using Animal and Dog where sound() displays "Dog barks". [Polymorphism – method overriding (sound())]
- 9) a) A Student class must be placed in a named group called college and accessed from a class belonging to another package. Identify the Java feature involved and write the program using college.Student and test.TestStudent.[Packages – college.Student / test.TestStudent]
b) A calculator should perform addition when either two or three integers are supplied, while keeping the same method name. Identify the Java concept involved and write a small program using overloaded add() methods. [Method overloading]

- 10) a) A BabyDog should be able to use behaviors available through both Dog and Animal. Identify the type of inheritance involved and write a small program using Animal2, Dog2 and BabyDog. . [Multilevel inheritance (Animal2→Dog2→BabyDog)]
- b) A child class needs to replace a method of its parent class but also execute the parent's version before its own version. Identify the Java feature involved and write a small program using Animal5 and Doggie. [Method overriding]
- 11) a) A general Shape should specify that every shape must calculate its area, while a Rectangle should provide the actual calculation. Identify the Java concept involved and write a small program using Shape and Rectangle. [Abstract class (Shape/Rectangle)]
- b) Different classes must follow a common rule containing a method that they must implement. Identify the Java feature involved and write a small program using interface A and class B. [Interface (A/B)]
- 12) a) A Student class needs to be organized under college and reused by another class under test. Identify the Java feature involved and write a small program showing how Student is accessed. [[Packages (college.Student, test)]
- b) A class should be prevented from being inherited by another class. Identify the Java keyword involved and write a small program using Bike and Honda1 to demonstrate it. [final keyword (Bike/Honda1)]

UNIT - 3

- 1) What is Exception Handling in Java? Explain try, catch, finally and throw with a suitable program.
- 2) Explain the exception handling mechanism in Java. Describe the roles of try, catch, throw, throws and finally with a suitable program.
- 3) What is an exception? Explain the concept of exception handling and its benefits in Java.
- 4) Differentiate between the **Termination model** and the **Resumptive model** of exception handling.
- 5) Explain the **exception hierarchy** in Java with a neat diagram.
- 6) What are built-in exceptions? Explain any five with examples.
- 7) How do you create your own (user-defined/custom) exception subclass? Write a program to demonstrate it.
- 8) Explain multiple catch blocks and nested try-catch statements with an example program.
- 9) A program attempts to divide 30 by 0. Instead of allowing the program to terminate abnormally, a user-friendly message should be displayed. Identify the Java mechanism involved and write the program using try and catch. [Exception handling – divide by zero (try/catch).]
- 10) A program may encounter an invalid operation during execution. The error should be caught and an appropriate message should be displayed instead of terminating abruptly. Identify the Java mechanism involved and write a small program to handle division by zero. [Exception handling — divide by zero.]